

Hourly electricity demand and temperature data from South Africa used for probabilistic forecasting with additive quantile regression models

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Abstract

This article presents a comprehensive description of the hourly electricity demand and temperature data used in developing probabilistic short-term load forecasting models. The data, sourced from a South African power utility company and the South African Weather Services, spans three years from January 2010 to December 2012. The dataset includes key variables such as hourly load, temperature readings from multiple weather stations, calendar effects (hour, month, day type, holidays), and derived variables like a nonlinear trend and lagged demand terms. This data descriptor details the data sources, preprocessing steps, variable selection using hierarchical Lasso, and the final structure of the dataset employed for modelling with additive quantile regression (AQR) and generalised additive models (GAMs), as well as for forecast combination using quantile regression averaging (QRA).

1 Specifications Table

Subject area	Energy Engineering and Power Technology
More specific subject area	Load Forecasting, Probabilistic Forecasting, Energy Systems
Type of data	Tables, Figures, Processed Excel/CSV files
How data was acquired	Provided by Eskom (South African power utility) and the South African Weather Service.
Data format	Filtered, cleaned, and analysed.
Experimental factors	Data cleaning involved handling missing values in temperature data via imputation. Variable selection was performed using the least absolute shrinkage and selection operator (Lasso) with hierarchical interactions to ensure model interpretability. Residual autocorrelation was addressed in the modelling framework.
Experimental features	The data was used to train and test four main models: GAM and AQR, each with and without pairwise interactions. Forecasts from these models were then combined using convex combination and QRA. Models were evaluated on their probabilistic forecasting accuracy, including prediction interval coverage and width.
Data source location	South Africa. Hourly load data is from the national grid. Temperature data is aggregated from 28 meteorological stations across the country.
Data accessibility	The data is hosted on a public repository, GitHub and is freely available to the public for non-commercial research purposes. Repository: https://github.com/csigauke/Probabilistic-Hourly-Load-Forecasting-Using-Additive-Quantile-Regression-M
Related research article	The relevant research article is: Probabilistic Hourly Load Forecasting Using Additive Quantile Regression Models. Energies, https://doi.org/10.3390/en11092208

2 Value of the Data

- The data provides a detailed, multi-year (2010-2012) time series of national hourly electricity load and corresponding regional temperature data for South Africa, enabling researchers to develop and validate both point and probabilistic load forecasting models.
- The dataset is valuable for exploring advanced statistical and machine learning techniques, particularly those that capture intra-day and seasonal patterns, such as additive models and quantile regression, which are crucial for understanding the full distribution of future load.
- By documenting the variable selection process using hierarchical Lasso, the data allows researchers to study the importance of main effects and pairwise interactions in load forecasting, including the nonlinear trend and lagged effects.
- The data facilitates research into forecast combination methods for probabilistic forecasting, such as quantile regression averaging (QRA), which can lead to more robust and accurate prediction intervals for grid management and unit commitment.

3 Data Description

The dataset described in this article consists of aggregated hourly electricity demand for South Africa and corresponding temperature data. The data covers the period from 1 January 2010 to 31 December

2012, comprising 26,281 observations. The primary response variable is hourly electricity load, while the predictor set includes calendar variables, temperature-derived variables, and lagged effects.

3.1 Variables

The dataset includes the following variables:

- **Load:** Hourly electricity demand (MW), the target variable.
- **Temperature Variables:** Processed from 28 weather stations. South Africa was divided into two main thermal regions (coastal and inland) and variables were averaged to create:
 - ADTC, maxTC, minTC: Average, maximum, and minimum daily coastal temperatures.
 - ADTI, maxTI, minTI: Average, maximum, and minimum daily interior temperatures.
 - AADTCI, AmaxTCI, AminTCI: Averages of coastal and interior daily, maximum, and minimum temperatures.
 - DADTCI, DmaxTCI, DminTCI: Differences between coastal and interior daily, maximum, and minimum temperatures.
- **Calendar and Holiday Variables:**
 - Hour: Factor variable for the hour of the day (1 to 24).
 - Month: Factor variable for the month of the year (1 to 12).
 - Daytype: Factor variable for the day of the week (1 = Monday, ..., 7 = Sunday).
 - Holiday: Binary indicator (1 for public holidays and the days immediately before and after a holiday).
- **Derived Trend and Lagged Variables:**
 - noltrend: A non-linear trend variable, extracted by fitting a penalised cubic regression spline to the load data to capture long-term patterns.
 - Lag1, Lag12, Lag24: Lagged values of hourly load at 1, 12, and 24-hour intervals to capture recent demand and daily cycles.
- **Interaction Variables:** Pairwise interactions between key predictors, selected using a hierarchical Lasso procedure, were included in some models. Examples for peak hours (e.g., 18:00, 19:00, 20:00) included interactions like ‘AED * DmaxTCI’ (average daily demand * difference in max temperatures) and ‘noltrend * Lag1’.

4 Experimental Design, Materials, and Methods

4.1 Data Sources and Preprocessing

The data were acquired from two primary sources: Eskom, the South African power utility, provided the aggregated hourly electricity load data. The South African Weather Service provided hourly temperature data from 28 meteorological stations across the country.

During the data cleaning phase, stations with excessive missing values were removed. For the remaining stations, missing temperature values were imputed using the ‘mice’ (Multivariate Imputation by Chained Equations) package in R. Following imputation, temperatures were averaged to create regional variables for the two main thermal regions in South Africa: coastal and inland. From these regional averages, several derived variables (e.g., differences and averages) were created as listed in Section 3.

4.2 Data Partitioning

The full dataset of 26,281 hourly observations was chronologically split into a training set and a testing set.

- **Training set:** 1 January 2010 to 2 April 2012 (19,708 observations, 75% of the data). This set was used for model estimation and variable selection.
- **Testing set:** 2 April 2012 to 31 December 2012 (6,573 observations, 25% of the data). This out-of-sample period was used to evaluate the forecasting performance of all developed models.

5 Acknowledgments

The authors acknowledge Eskom, South Africa's power utility, and the South African Weather Service for providing the data used in this study.

6 Funding sources

This research was funded by the National Research Foundation of South Africa (Grant Number: 93613.).

References

- [1] Eskom Data Portal <https://www.eskom.co.za/dataportal/>
- [2] Sigauke, C.; Nemukula, M.M.; Maposa, D. Probabilistic Hourly Load Forecasting Using Additive Quantile Regression Models. *Energies* 2018, 11, 2208. <https://doi.org/10.3390/en11092208>

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